

Arsenic, Organic Foods, and Brown Rice Syrup



Background: Rice can be a major source of inorganic arsenic (Asi) for many sub-populations. Rice products are also used as ingredients in prepared foods, some of which may not be obviously rice based. Organic brown rice syrup (OBRS) is used as a sweetener in organic food products as an alternative to high-fructose corn syrup. We hypothesized that OBRS introduces As into these products. Objective: We determined the concentration and speciation of As in commercially available brown rice syrups and in products containing OBRS, including toddler formula, cereal/energy bars, and high-energy foods used by endurance athletes. Methods: We used inductively coupled plasma mass spectrometry (ICP-MS) and ion chromatography coupled to ICP-MS to determine total As (Astotal) concentrations and As speciation in products purchased via the Internet or in stores in the Hanover, New Hampshire, area. Discussion: We found that OBRS can contain high concentrations of Asi and dimethyl-arsenate (DMA). An “organic” toddler milk formula containing OBRS as the primary ingredient had Astotal concentrations up to six times the U.S. Environmental Protection Agency safe drinking water limit. Cereal bars and high-energy foods containing OBRS also had higher As concentrations than equivalent products that did not contain OBRS. Asi was the main As species in most food products tested in this study. Conclusions: There are currently no U.S. regulations applicable to As in food, but our findings suggest that the OBRS products we evaluated may introduce significant concentrations of Asi into an individual’s diet. Thus, we conclude that there is an urgent need for regulatory limits on As in food.